

Dexterous Manipulation with Deep Reinforcement Learning: Efficient, General, and Low-Cost

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Abstract—Dexterous multi-fingered robotic hands can perform a wide range of manipulation skills, making them an appealing component for general-purpose robotic manipulators. However, such hands pose a major challenge for autonomous control, due to the high dimensionality of their configuration space and complex intermittent contact interactions. In this work, we propose deep reinforcement learning (deep RL) as a scalable solution for learning complex, contact rich behaviors with multi-fingered hands. Deep RL provides an end-to-end approach to directly map sensor readings to actions, without the need for task specific models or policy classes. We show that contact-rich manipulation behavior with multi-fingered hands can be learned by directly training with model-free deep RL algorithms in the real world, with minimal additional assumption and without the aid of simulation. We learn to perform a variety of tasks on two different low-cost hardware platforms entirely from scratch, and further study how the learning can be accelerated by using a small number of human demonstrations. Our experiments demonstrate that complex multi-fingered manipulation skills can be learned in the real world in about 4-7 hours for most tasks, and that demonstrations can decrease this to 2-3 hours, indicating that direct deep RL training in the real world is a viable and practical alternative to simulation and model-based control. <https://sites.google.com/view/deeprl-handmanipulation>

I. INTRODUCTION

A long standing goal in robotics is to create general purpose robotic systems that operate in a wide variety of human-centric environments such as homes and hospitals. For robotic agents to be competent in such unstructured environments, versatile manipulators like multi-fingered hands are needed in order to cope with the diversity of tasks presented in human-centric settings. However, the versatility of multi-fingered robotic hands comes at the price of high dimensional configuration spaces and complex finger-object contact heavy interactions, making modeling and controller synthesis particularly challenging.

Reinforcement learning (RL) and optimal control techniques provide generic paradigms for optimizing complex controllers that are hard to specify by hand. In particular, model-free RL provides a way to optimize controllers end-to-end without any explicit modeling or system identification. This reduces human engineering effort and produces controllers that are directly adapted to the physical environment, which makes for a scalable and general approach to robotic manipulation. Such systems do not require an explicit model



Fig. 1: We demonstrate that deep RL can learn a wide range of dexterous manipulation skills with multi-fingered hands, such as opening door with flexible handle, rotating a cross-shaped valve, rotating the same valve made of deformable foam, and box flipping.

of the robot, calibration of the sensors, or other manual description that is typically needed for model-based methods.

In this work, we study how model-free RL can be scaled up to learn a variety of manipulation behaviors with multi-fingered hands directly in the real world, using general-purpose neural network policy representations without manual controller or policy class design. We conduct our experiments with low cost multi-fingered manipulators, and show results on tasks such as rotating a valve, flipping a block, and door opening. Somewhat surprisingly, we find that successful neural network controllers for each of these tasks can be trained directly in the real world in about 4-7 hours for most tasks.

We also show that we can further accelerate the learning process by incorporating a small number of human demonstrations, building on the recently proposed DAPG [1] algorithm. Using only 20 demonstrations, obtained through kinesthetic teaching, the learning time can be brought down to around 2–3 hours which corresponds to a 2x speedup. Together, these results establish that model-free deep RL and demonstration-driven acceleration provide a viable approach for real-world learning of diverse manipulation skills.

The main contribution of this work is to demonstrate that model-free deep reinforcement learning can learn contact

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rich manipulation behaviors on low-cost multi-fingered hands directly in the real world. We demonstrate this on two different low cost robotic hands, performing three distinct tasks on both rigid and deformable objects. This result is particularly relevant in scenarios which are hard to simulate and real world training is the only feasible option. We further demonstrate that a small number of demonstrations can accelerate learning and analyze a number of design choices and performance parameters.

II. RELATED WORK

In recent years, multi-fingered robotic hands have become more and more widespread as robotic manipulation has progressed towards more challenging tasks. A significant portion of this research has focused on the physical design of anthropomorphic hands, and designing simple controllers for these hands. The majority of these manipulators [2], [3] are custom, expensive, heavily instrumented, and fragile. In contrast, the results presented in this work are on low-cost (and relatively robust) commodity hardware, with limited sensing and actuation capabilities. As our method trains directly in the real world, the resulting solution encapsulates the sensory and control inaccuracies that are associated with such hardware thereby making them more deployable.

Much of the work on robotic hands has focused on grasping [4], as compared to other dexterous skills [5]. Such work is often focused on achieving stable grasps by explicitly reasoning about the stability of the grasp using geometric and analytic metrics [6], [7], [8]. While considerable progress has been made on the theoretical side [9], sensing and estimation challenges have limited their applicability in the real world considerably. In this work, we consider tasks which require significant finger coordination and dexterity, making the manual design of controllers very challenging. Our proposed solution using model-free deep RL alleviates the need for manual controller design or modeling.

In simulation, [10], [11], [12] have shown the ability to synthesize complex behaviors when provided with a ground truth model. This model is rarely available in the real world. Techniques such as [13], [14] get around this problem by learning locally-linear models and deploying model-based RL method to learn instance specific linear controllers demonstrating turning rods and arranging with beads. [15] shows the ability to learn in-hand repositioning in the real world, with a reinforcement learning method (NPRESS) and non-parametric control policies. Similar methods have also been used for whole-arm manipulation [16]. While these methods train in the real world, the resulting policies involve relatively simple individual motions. In contrast to these model-free experiments, our results show that deep RL can learn complex contact-rich behaviors with finger gaits, including with combined hand and arm control. In contrast to the model-based methods, we show that this can be done directly in the real world without any model.

We also show how the learning process can be accelerated using a small number of human demonstrations. Accelerating reinforcement learning with demonstrations has been the

subject of a number of works in the reinforcement learning community [17], [16], [18], [19], [1], [20], [21] but these methods have not been applied to real world manipulation with multi-fingered hands. We show that this is indeed possible and very effective using algorithms which combine behavior cloning and policy gradient. We build on the algorithm proposed by us in prior work [1], and show that this approach can indeed scale to real world dexterous manipulation, with significant acceleration benefits.

Another related line of research seeks to transfer policies trained in simulation into the real world. Aside from rigorous system identification, a recent class of methods has focused on randomizing the simulation, both for physical [22] and visual [23] transfer. This approach has been employed for manipulation [24], visual servoing [25], locomotion [26], and navigation [23]. Recent concurrent work has also applied this approach to multi-fingered hands for an object rotation task [27] with about 50 hours of computation, equivalent to 100 years of experience. In contrast, our approach trains directly in the real world in a few hours without the need for manual modeling. We discuss the relative merits of real-world and simulated training in Section VI-D.

III. HARDWARE SETUP

In order to demonstrate the generalizable nature of the model-free deep RL algorithms, we consider two different hardware platforms: a custom built 3-fingered hand, referred to as the Dynamixel claw (Dclaw), and a 4-fingered Allegro hand. Both hands are relatively cheap, especially the Dclaw. For several experiments, we also mounted the hands on a Sawyer robot arm.

a) Dynamixel Claw: The Dynamixel claw (Dclaw) is custom built using Dynamixel servo motors, which are low-latency, position controlled and equipped with angle sensing. It is a 3-fingered, 9 degrees of freedom manipulator which costs under \$2,500 to construct. Dclaw is robust and is able to run more than 24 hours without intervention or hardware damage.

b) Allegro Hand: The Allegro hand is a 4-fingered anthropomorphic hand, with 16 degrees of freedom, and can handle payloads of up to 5 kg. This hand uses DC motors for manipulation and can be either torque or position actuated using a low level PID controller. The Allegro hand costs on the order of \$15,000.

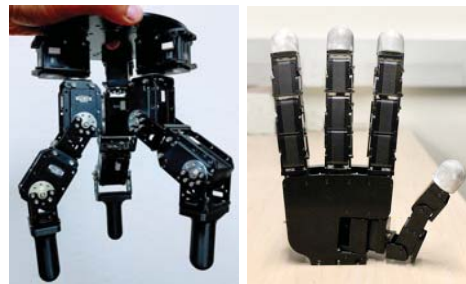


Fig. 2: Left: 3 finger Dynamixel claw. Right: 4 finger anthropomorphic Allegro hand

IV. TASKS

While the approach we describe for learning dexterous manipulation is general and broadly applicable, we consider three distinct tasks in our experimental evaluation - valve rotation, box flipping, and door opening. These tasks involve challenging contact patterns and coordination, and are inspired by everyday manipulation tasks. We approach these problems using RL where the goal is to learn a control policy which maximizes a user-provided reward function, which we define independently for each of our tasks below.

a) *Valve Rotation*: This task involves turning a valve or faucet to a target position. The fingers must cooperatively push and move out of the way, posing an exploration challenge. Furthermore the contact with the valve complicates the dynamics. For our task, the valve must be rotated from 0° to 180° .



Fig. 3: Illustration of valve rotation

The state space consists of all the joint angles of the hand, the current angle of rotation of the valve, the goal angle, and the last action taken. The action space is the joint angles of the hand and the reward function is

$$r = -d\theta + 10 * \mathbb{1}_{\{d\theta < 0.1\}} + 50 * \mathbb{1}_{\{d\theta < 0.05\}}$$

$$d\theta := |\theta_{\text{valve}} - \theta_{\text{goal}}|$$

We define a trajectory as a success if $d\theta < 20^\circ$ for at least 20% of the trajectory.

b) *Vertical box flipping*: This task involves rotating a rectangular box, which freely spins about its long axis, from 0° to 180° . This task also involves learning coordinated motions of the fingers such that while the top finger is pushing, the bottom two move out of the way, and vice versa.



Fig. 4: Illustration of box flipping

The state space consists of all the joint angles of the hand, the current angle of rotation of the box, the goal angle, and the last action taken. The action space consists of the joint angles of the hand and the reward function and success metric are identical to those defined for valve rotation.

c) *Door opening*: This task involves both the arm and the hand working in tandem to open a door. The robot must learn to approach the door, grip the handle, and then pull backwards. This task has more degrees of freedom given the additional arm, and involves the sequence of actions: going to the door, gripping the door, and then pulling away.

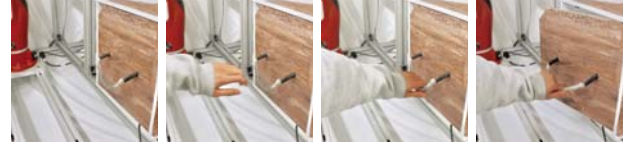


Fig. 5: Illustration of door opening.

The state space is all the joint angles of the hand, the Cartesian position of the arm, the current angle of the door, and last action taken. The action space is the joint angles of the hand and horizontal 2D position of the wrist of the arm. The reward function is provided as

$$r = -(d\theta^2 + dx)$$

$$d\theta := |\theta_{\text{door}} - \theta_{\text{closed}}|, dx = |x_{\text{arm}} - x_{\text{door}}|$$

We define a trajectory as a success if at any point $d\theta > 30^\circ$.

A. State Estimation with Encoder Sensors

For these tasks, we solve both the problem of state estimation and resetting using a setup with objects augmented with Dynamixel servo motors. These motors serve the dual purpose of resetting objects (such as the valve, box, or door) to their original positions and measuring the state of the object ($\theta_{\text{valve}}, \theta_{\text{box}}, \theta_{\text{door}}$).

V. ALGORITHMS

In this work, we show that the tasks described in the previous section can be solved using model-free on-policy reinforcement learning algorithms. This requires 7 hours for valve turning, 4 for box flipping and 16 hours for door opening. We also demonstrate that the learning time can be significantly reduced by using a small number of human demonstrations.

A. Policy Gradient

Reinforcement learning algorithms operate within the framework of Markov decision processes. The goal in reinforcement learning is to find a policy π that describes how to choose actions in any given state, such that we maximize the sum of expected rewards.

In this work, we parameterize the policy using a neural network, and use a natural policy gradient (NPG) ascent based approach. Natural policy gradient methods capture curvature information about the optimization landscape, thereby stabilizing the optimization process and enabling faster convergence. There are numerous variants of the natural policy gradient method [28], [29]. For simplicity, we use the truncated natural policy gradient as described in [30]. Please refer to [1] for more details.

B. Demonstration Augmented Policy Gradient

While the NPG algorithm is guaranteed to converge asymptotically to at least a locally optimal policy, the rate of convergence could be very slow. In such cases, we would like to accelerate the learning process by using various forms of prior knowledge.

One way to incorporate human priors is through the use of demonstration data, obtained through say kinesthetic teaching.

In [1], we developed the demonstration augmented policy gradient (DAPG) algorithm which combines reinforcement learning with imitation learning using the demonstration data. DAPG starts by pre-training the policy with behavior cloning on demonstration data and subsequently finetuning it using an augmented policy gradient.

This DAPG algorithm has been shown to significantly accelerate the learning by improving exploration, and exceed the performance of the demonstrations while still retaining their stylistic aspects – all in simulation. In this work, we demonstrate that this algorithm provides a practical and useful way of accelerating deep RL on real hardware to solve challenging manipulation problems.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

The goal of our experiments is to empirically address the following research questions:

- Can model-free deep RL provide a practical means for learning various dexterous manipulation behaviors directly in the real world?
- Can model-free deep RL algorithms learn on different hardware platforms and on different physical setups?
- Can we accelerate the learning process using a small number of demonstrations obtained through kinesthetic teaching?
- How do particular design choices of the reward function and actuation space affect learning?

To do so, we utilize the tasks in Section IV and algorithms described in Section V. Additional experimental details and videos can be found at the supplementary website (<https://sites.google.com/view/deeprl-handmanipulation>).

A. Model-Free Deep RL

First, we explore the performance of model-free deep reinforcement learning on our suite of hardware tasks. The learning progress is depicted in Fig 9 and also in the accompanying video. We find that, somewhat surprisingly, model-free deep RL can acquire coherent manipulation skills in the time scales of a few hours (7 hours for turning a valve, 4 hours to flip a box, 16 hours for opening a door). These training times reflect the time taken till the policy achieves 100% success rate over 10 evaluation rollouts, according to the success metrics defined in Section IV (Fig 14). The algorithm is robust and did not require extensive hyperparameter optimization. The only hyperparameter that was tuned was the variance of the initial policy for exploration. We analyze specific design choices in the reward function and actuation scheme in Section VI-E.



Fig. 6: Visualization of a policy that has learned to turn a valve after 7 hours of training.



Fig. 7: Visualization of a policy that has learned to flip a box after 4 hours of training. The Dclaw learns to extend its bottom two fingers and then push its top finger forwards.



Fig. 8: Visualization of a policy that has learned to open a door after 16 hours of training. The robot learns to move towards the door, grasp the deformable handle, and then pull the door open.

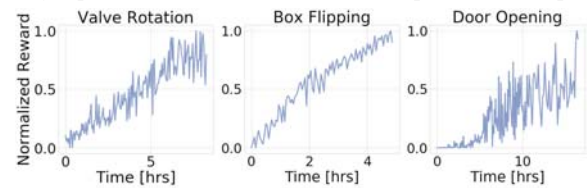


Fig. 9: Learning progress with model-free RL from scratch using the NPG algorithm. The policies reach 100% average success rates after 7.4 hours for valve rotation and 4.1 hours for box flipping and 15.6 hours for door opening. The performances have been normalized such that a random policy achieves a score of 0 and the best policy gets a score of 1. The final policies all get over 90% success rate on the metrics defined in Section IV

We find that for the valve and the box flipping, the learning monotonically improves on the continuous reward signal, whereas for opening the door the learning is more challenging, since high reward is only obtained when the door is actually opened. The agent has to consistently pull open the door in order to see the reward, leading to the large spikes in learning as seen in Fig. 9.

The learned finger gaits that we observe have interesting coordination patterns. The tasks require that the fingers move nimbly and with coordination so as to rotate the objects and then move out of the way. This behavior can be appreciated in the accompanying video on the supplementary website.

B. Learning on Different Hardware and Different Materials

To illustrate the ability of model-free RL algorithms to be applied easily to new scenarios and robots, we evaluated the valve task using the exact same deep RL algorithm with a different hand – the 4-fingered, anthropomorphic Allegro hand. We find that the Allegro hand was able to learn this task, as illustrated in Fig. 12.



Fig. 10: The Allegro hand learns to rotate a valve 180°.

Both systems are able to quickly learn the right behavior

with the same hyperparameters. While morphology does indeed have an effect on rate of learning, we find that both systems eventually succeed at the task. The easy adaptability of these algorithms is extremely important. We don't need to construct an accurate simulation or model of each new robot; instead we can simply run the same algorithm repeatedly to learn new behaviors.

Besides changing the robot's morphology, we can also modify the object that is manipulated. We evaluated learning with a different valve material, namely a soft foam (Fig. 11). The contact dynamics with such a deformable material are hard to simulate and the claw can deform the valve in many different directions, making the actual manipulation task challenging. We see that model-free RL is able to learn to manipulate with the foam valve effectively, even generating behaviors that exploit the deformable structure of the object.



Fig. 11: Dclaw learns to rotate a foam valve despite its deformable structure. The claw learns to focus its manipulation on the rigid center of the valve.

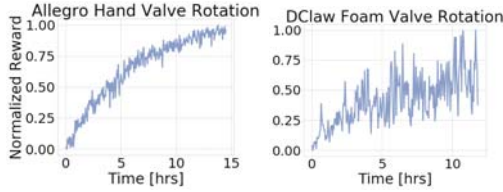


Fig. 12: Left: learning progress of Allegro hand rotating rigid screw. Right: learning progress of Dclaw rotating the foam valve. The rewards have been normalized such that a random policy achieves score of 0 and the final trained policy achieves a score of 1.

C. Accelerating Learning with Demonstrations

While we find that model-free deep RL is generally practical in the real world, the number of samples can be further reduced by using demonstrations, as discussed in Section V-B. In order to understand role of demonstrations, we collected 20 demonstrations per task via kinesthetic teaching.

The demonstrations are suboptimal but can still provide guidance for exploration and accelerate the learning process. With only a few demonstrations, we see that demonstration augmented policy gradient (DAPG) can speed up learning significantly, dropping learning time by 2x (Fig. 13). We record training times across tasks using the previously defined success metrics in (Fig. 14).

The efficiency of DAPG comes from the fact that the behavior cloning initialization gives the agent a rough idea of how to solve the task, and the augmented loss function guides learning through several iterations. The behaviors learned are also more gentle and legible to humans than behaviors learned via training from scratch, which is showcased in the accompanying video. To better understand the learned

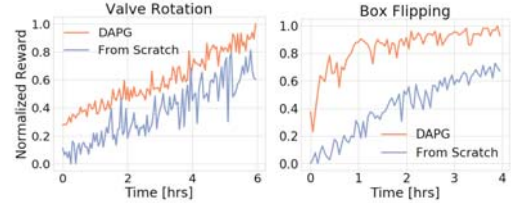


Fig. 13: Learning progress with training from scratch using NPG and using DAPG. The performances have been normalized such that a random policy achieves a score of 0 and the best DAPG policy gets a score of 1. The final policies all get over 90% success rate on the metrics defined in Section IV

behaviors, we also evaluated the robustness of these behaviors to variations in the initial position of the valve, and to noise injected into actions and observations (Fig. 15).

Fig. 14: Training Times Across Tasks [hrs]. Training time is determined using the success metrics defined in Section IV. A training run is complete once the deterministic policy achieves 100% success rate over 10 evaluation rollouts.

Task	From Scratch	DAPG
Valve	7.4	3.0
Box	4.1	1.5
Door	15.6	—



Fig. 15: Plots showing robustness of learning with DAPG vs from scratch for valve rotation. Left: Success rate with change in valve initial position. Right: Success rate with x% uniformly random noise injected into observations and actions. DAPG is more robust with change in initial valve position, but less robust to noise.

We also collect demonstrations from a distribution of valve screwing tasks and use DAPG to generalize to a task sampled from the demo distribution. Unsurprisingly, it is not as effective as collecting demonstrations that directly solve that task, but is able to learn faster than from scratch (Fig. 16).

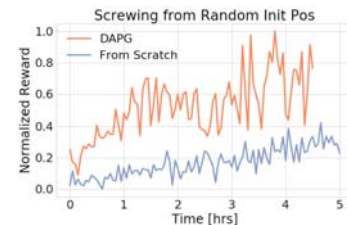


Fig. 16: Training Dclaw to turn a valve from a single randomly sampled initial position between $[-45^\circ, 45^\circ]$ to 180° . DAPG was trained with 20 demos that turned the valve from initial positions sampled uniformly at random between $[-45^\circ, 45^\circ]$ to 180° .

D. Performance with Simulated Training

While the main goal of this work is to study how model-free RL can be used to learn complex policies directly on real hardware, we also evaluated training in simulation and transfer, employing randomization to allow for transfer [22], [23]. This requires modeling the task in a simulator and manually choosing the parameters to randomize. We see in Fig. 17 that the randomization of PID parameters and friction is crucial for effective transfer.

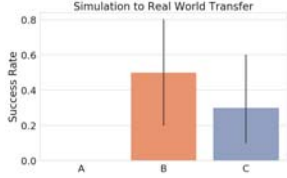


Fig. 17: Success rates using different sim2real transfer strategies for valve turning. A: No domain randomization. B: Randomization of position control PID parameters and friction, without previous action in state. C: Same as B, but with previous action in state.

While simulation to real transfer enabled by randomization is an appealing option, especially for fragile robots, it has a number of limitations. First, the resulting policies can end up being overly conservative due to the randomization, as widely observed in robust control. Second, the particular choice of parameters to randomize is crucial for good results, and insights from one problem domain may not transfer to others. Third, increasing randomization results in more complex models tremendously increasing the training time and required computational resources, as discussed in Section II. Finally, and perhaps most importantly, an accurate simulator must be constructed manually, with each new task modeled by hand in the simulation, which requires substantial time and expertise. For tasks such as valve rotation with the foam valve or door opening with a soft handle, creating the simulation itself is very challenging. Thus training in the real world may be more efficient and lead to better policies.

E. Design Choices

To understand design choices needed to train dexterous hand manipulation systems, we analyzed different factors which contribute to learning. We performed this analysis in simulation to choose schemes for real world training.

1) *Choices of Action Space:* The choice of actuation space often makes an impact on learning progress. For hand manipulation it also greatly affects the smoothness of rollouts, and hence sustainability of the hardware. We consider a comparison between using position and torque controllers as well as their higher order derivatives. We compare the vibrations induced by each of these control schemes by measuring the sum of magnitudes of the highest half of Fourier coefficients of joint angles from sample trajectories induced by random actions (Fig. 18).

We see that position control has the lowest vibration amongst all the choices of control schemes, and is also able to achieve significantly better performance. This is likely because we are using a stabilizing PID control at the low

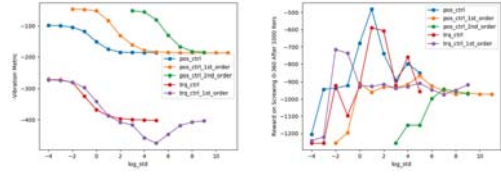


Fig. 18: Left: Analysis of vibrations induced by different actuation schemes. Higher metric indicates lower vibrations. We find that position control is able to induce significantly lower vibrations than torque control, making it safer to run on hardware. Right: Analysis of rewards using the control scheme on valve rotating task.

level to do position control which reduces the load on the learning algorithm. It is also harder to learn a policy when controlling higher order derivatives, and it is easier to learn with position control than with torque control.

2) *Impact of Reward Function:* We also investigated the effect of the reward function on learning progress. To provide some intuition about the different choice of reward functions, we show a comparison between 3 different reward functions for learning. We evaluate learning progress for these three reward functions in simulation, to choose an appropriate form of reward for real world training.

- 1) $r_1 = -\|\theta - \theta_{goal}\|_2$
- 2) $r_2 = r_1 + 10 * \mathbb{1}_{\{r_1 < 0.1\}} + 50 * \mathbb{1}_{\{r_1 < 0.05\}}$
- 3) $r_3 = r_2 - \|v\|_2$

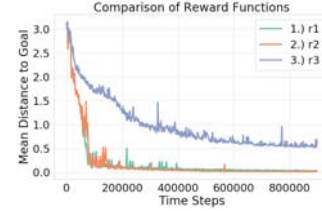


Fig. 19: Analysis of learning progress of valve rotation in simulation with different reward functions.

We find that learning is most effective with using either option 1 or 2. The control cost ensures smoother operation, but at the cost of efficiency in learning, since optimization of control penalties results in reduction of exploration. In real world experiments we found that training without a control cost still produced safe behaviors.

VII. DISCUSSION AND FUTURE WORK

In this work, we study real-world model-free reinforcement learning as a practical, efficient, and general means to learn complex dexterous manipulation behaviors on high dimensional multi-fingered hands. Model-free RL algorithms can easily be applied to a variety of low-cost hands, and solve a variety of challenging tasks that are hard to simulate accurately. This kind of generality and minimal manual engineering may be a key ingredient in endowing robots with the kinds of large skill repertoires. We also show that the sample complexity of model-free RL can be substantially reduced with suboptimal kinesthetic demonstrations. There are several exciting avenues for future work. Studying dexterous manipulation with visual observations is promising. Furthermore, an exciting direction is to pool experience across multiple skills and study the multi-task setting.

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